Due on monday, December 5th, in class

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1. Consider a single particle in one dimension.

   (a) **[1 pt.]** If the position operator $\hat{x}$ is applied to a wavefunction $\phi(x)$, what is the result?

   (b) **[2 pts.]** If the momentum operator $\hat{p}$ is applied to a wavefunction $\phi(x)$, what is the result?

   (c) **[5 pts.]** Apply the commutator $[\hat{x}, \hat{p}]$ on an arbitrary wavefunction (arbitrary function of x). Hence evaluate the operator $[\hat{x}, \hat{p}]$.

   (d) **[6 pts.]** Define the operator $\hat{M} \equiv \frac{d^2}{dx^2} \equiv \partial_x^2$. (As usual I am sloppy about distinguishing full and partial derivatives.) By applying on an arbitrary function of x, evaluate the commutator $[\hat{x}, \hat{M}]$.

2. The Hamiltonian of a system has energy eigenvalues ϵ_j and the corresponding orthornormal eigenstates are $|\phi_j\rangle$. The system is prepared in the initial state

$$|\psi(0)\rangle = \frac{1}{2}|\phi_2\rangle + \frac{\sqrt{3}}{2}|\phi_5\rangle$$

   Here $|\psi(t)\rangle$ represents the wavefunction at time t.

   Note: The system is defined completely generally. Do not assume a particular dimension of the Hilbert space. The system could be any quantum system - one particle in three dimensions, a spin-1/2 object, a collection of fifty spin-1/2 objects, twenty particles in two dimensions,.... etc. It should not be necessary or even useful to assume a particular type of system. E.g., do not assume inner products to involve integrals over a single spatial coordinate or that states/kets are vectors of dimension 3.

   (a) **[2 pts.]** If the energy of the system is measured, what are the possible results? With what probabilities will these values be found?

   (b) **[2 pts.]** Assume in the following that no measurement is performed at $t = 0$. Instead, the system is allowed to evolve starting from the state $|\psi(0)\rangle$ given above.

   What is the state of the system, $|\psi(t)\rangle$, at a later time t?

(c) **[7 pts.]** The operator $\hat{B}$ has zero expectation value in each of the eigenstates, i.e., $\langle \phi_j | \hat{B} | \phi_j \rangle = 0$ for all j. In addition, the so-called off-diagonal matrix elements are all equal and real:

$$\langle \phi_i | \hat{B} | \phi_j \rangle \; = \; \beta \quad \text{whenever} \quad i \neq j.$$

Calculate $\langle \hat{B} \rangle$ as a function of time in the state $| \psi(t) \rangle$, i.e., calculate

$$\langle \hat{B}(t) \rangle \; = \; \langle \psi(t) | \hat{B} | \psi(t) \rangle$$

Describe in a sentence the time-dependence of the expectation value of the observable B. If there is a frequency involved, what is it?

(d) **[2 pts.]** Why are $\langle \phi_i | \hat{B} | \phi_j \rangle$ called the off-diagonal matrix elelments whenever $i \neq j$?

(e) **[4 pts.]** A measurement of energy is performed at time $t = t_0$ and yields the value ϵ_5. What is the state of the system immediately after this measurement? What is the state of the system at a later time $t > t_0$

(f) **[3 pts.]** What will be the behaviour of $\langle \hat{B} \rangle$ after the measurement, at times $t > t_0$? Justify/explain your answer.

(g) **[3 pts.]** Sketch a possible plot of $\langle \hat{B} \rangle$ versus time, running from $t = 0$ to $t = 2t_0$.

3. **[5 pts.]** Show that the eigenvalues of a unitary operator have unit modulus, i.e., that any eigenvalue of a unitary operator is a pure phase $e^{i\lambda}$.

4. The state $| \mu_1 \rangle$ is an eigenstate of $\hat{M}$ with corresponding eigenvalue m_1.

(a) **[3 pts.]** Show that $| \mu_1 \rangle$ is also an eigenstate of $\hat{M}^2$, and find the corresponding eigenvalue. Show that $| \mu_1 \rangle$ is also an eigenstate of $\hat{M}^n$, where n is an integer, and find the corresponding eigenvalue.

(b) **[5 pts.]** How is the operator $e^{\hat{M}}$ defined?
How is $e^{\alpha \hat{M}}$ defined, if α is a real number?
Is $| \mu_1 \rangle$ is an eigenstate of $e^{\alpha \hat{M}}$? If so, find the corresponding eigenvalue.